time, output noise is reduced.

S6 works as an on/off switch. The power supply is applied to the circuit through S6.

S5 works as a stereo/mono selector switch. If S5 is closed, the circuit works as a dual-channel mono amplifier. Left and right audio signals coming from connector CON1 are shorted through S5. If S5 is open, the circuit works as a stereo amplifier with two different input signals.

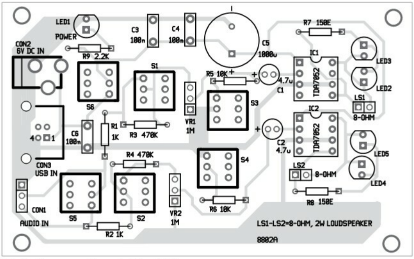

Fig. 3: Component layout of the PCB

Output power of TDA7052 is typically 1.2W over a load of 8-ohm and with power supply of 6V. The power supply can be increased but the load should have higher resistance, of say, 32-ohm.

The left and right audio channels have separate volume controls, VR1 and VR2, respectively. This avoids the usage of the balance-control potentiometer so that the two channels can also be used separately for the amplification of two mono signals.

Values of VR1 and VR2 should be minimum in order to reduce output noise. Value of C5 should be minimal for the circuit to operate from most USB interfaces.

Construction and testing

An actual-size PCB layout of the stereo amplifier with two TDA7052 amplifiers for portable devices is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable enclosure.

The circuit does not need any adjustment and will work immediately after proper assembly. LED2 and LED3 are visual indicators for left-channel audio output, while LED4 and LED5 are for right-channel audio output. EFY

Petre Tzv Petrov was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and expert-lecturer in OFPPT(Casablance), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria